SE444 Final Exam
Complete Formula Sheet & Study Guide

1. PROBABILITY FUNDAMENTALS

Probability Axioms

1. $\boxed{0 \leq P(A) \leq 1}$ for any event A
2. $\boxed{P(\Omega) = 1}$ (sample space)
3. $\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$

Complement Rule:

$$\boxed{P(\neg A) = 1 - P(A)}$$

Joint Probability:

$$\boxed{P(A, B) = P(A \cap B) = P(A \wedge B)}$$

Probability that both A and B occur together

Marginal Probability (Marginalization):

$$\boxed{P(A) = \sum_b P(A, B = b) = \sum_b P(A|B = b) \cdot P(B = b)}$$

Sum over all possible values of B to "integrate out" B

Conditional Probability:

$$\boxed{P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(A \cap B)}{P(B)}}$$

Probability of A given that B has occurred

Product Rule (Chain Rule):

$$\boxed{P(A, B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)}$$

$$\boxed{P(A, B, C) = P(A) \cdot P(B|A) \cdot P(C|A, B)}$$

BAYES' THEOREM (Most Important!)

$$\boxed{P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}}$$

Components:

- $P(H)$ = **Prior** probability of hypothesis
- $P(E|H)$ = **Likelihood** of evidence given H
- $P(E)$ = **Evidence** (normalizing constant)
- $P(H|E)$ = **Posterior** probability after seeing E

Total Probability Theorem:

$$\boxed{P(E) = \sum_i P(E|H_i) \cdot P(H_i)}$$

Expand $P(E)$ over all hypotheses H_i

Bayes with Binary Hypothesis:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E|H) \cdot P(H) + P(E|\neg H) \cdot P(\neg H)}$$

Independence:

$$A \perp B \iff \boxed{P(A, B) = P(A) \cdot P(B)}$$

$$A \perp B \iff P(A|B) = P(A)$$

Conditional Independence:

$$A \perp B \mid C \iff \boxed{P(A, B|C) = P(A|C) \cdot P(B|C)}$$

A and B are independent given C

2. BAYESIAN NETWORKS

Definition: A DAG (Directed Acyclic Graph) where:

- Each node = random variable
- Each edge = direct probabilistic dependency
- Each node has a CPT $P(X_i|\text{Parents}(X_i))$

BN Chain Rule (Factorization):

$$\boxed{P(X_1, ..., X_n) = \prod_{i=1}^{n} P(X_i|\text{Parents}(X_i))}$$

Joint distribution factorizes according to graph structure

Number of CPT Parameters:

$$\boxed{\text{params} = (d - 1) \cdot d^k}$$

d = values per variable, k = number of parents

d-SEPARATION (Conditional Independence)

Three Canonical Structures:

1. CHAIN: $A \to B \to C$

- Path **blocked** when B observed: $A \perp C \mid B$
- Path **active** when B not observed: $A \not\perp C$

2. FORK: $A \leftarrow B \to C$

- Path **blocked** when B observed: $A \perp C \mid B$
- Path **active** when B not observed: $A \not\perp C$

3. COLLIDER (V-Structure): $A \to B \leftarrow C$

- Path **blocked** when B NOT observed: $A \perp C$
- Path **OPENS** when B or descendant observed: $A \not\perp C \mid B$

Explaining Away Effect:

When collider B is observed, its parents become dependent. Observing one parent "explains away" the effect, reducing belief in the other parent.

EXACT INFERENCE

Inference by Enumeration:

$$P(Q|E=e) = \frac{\sum_h P(Q, H=h, E=e)}{\sum_{q,h} P(Q=q, H=h, E=e)}$$

Sum over all hidden variables H, normalize

Variable Elimination Steps:

1. Convert CPTs to factors $f_i(X_{i1}, ..., X_{ik})$
2. Restrict factors by evidence (set observed values)
3. For each hidden variable H to eliminate:
   - Multiply all factors containing H
   - Sum out H: $f'(X) = \sum_h f(X, H=h)$
4. Multiply remaining factors, normalize

Factor Operations:

- **Restrict:** Set variable to observed value
- **Multiply:** Pointwise product of factors
- **Sum out:** Marginalize over variable

3. DECISION THEORY

EXPECTED UTILITY (MEU Principle)

$$EU(a) = \sum_{s \in S} P(s) \cdot U(a, s)$$

Rational Agent: Choose action maximizing EU

$$a^* = \arg\max_a EU(a)$$

Decision Under Uncertainty:

- **States** S: Possible world outcomes
- **Actions** A: Choices available to agent
- **Utility** $U(a, s)$: Value of action a in state s
- **Probability** $P(s)$: Likelihood of state s

Value of Perfect Information (VPI):

$$VPI(E) = EU(\text{optimal with } E) - EU(\text{optimal without } E)$$

How much would you pay to know E before deciding?

MARKOV DECISION PROCESSES (MDPs)

MDP Definition: (S, A, P, R, γ)

- S = finite set of states
- A = finite set of actions
- $P(s'|s, a)$ = transition probability
- $R(s, a)$ or $R(s, a, s')$ = reward function
- $\gamma \in [0, 1)$ = discount factor

Discounting:

$$\text{Value} = R_0 + \gamma R_1 + \gamma^2 R_2 + ... = \sum_{t=0}^{\infty} \gamma^t R_t$$

$\gamma = 0$: myopic (immediate only); $\gamma \to 1$: farsighted

Value Function:

$$V^\pi(s) = E\left[\sum_{t=0}^{\infty} \gamma^t R_t \mid s_0 = s, \pi\right]$$

Expected cumulative discounted reward starting from s under policy π

BELLMAN EQUATION (Optimal Value)

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s')\right]$$

BELLMAN EQUATION (Q-Value)

$$Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q^*(s', a')$$

Value Iteration Algorithm:

1. Initialize $V_0(s) = 0$ for all s
2. Repeat until convergence ($|V_{k+1} - V_k| < \epsilon$):

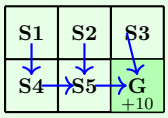
$$V_{k+1}(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V_k(s')\right]$$

Policy Extraction:

$$\pi^*(s) = \arg\max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s')\right]$$

Convergence: Guaranteed for $\gamma < 1$ (contraction mapping)

MDP Grid World Example:



$R = -1$ per step, $\gamma = 0.9$

Example Calculation for S5:

$$V(S5) = \max_a [R + \gamma \sum P \cdot V']$$

$$= -1 + 0.9 \times [0.8 \times 10 + 0.1 \times V(S2) + 0.1 \times V(S4)]$$

Blue arrows show optimal policy π^* pointing toward goal G

4. CLASSICAL PLANNING

STRIPS Representation

- **State:** Set of ground atoms (propositions that are true)
- **Goal:** Conjunction of ground atoms to achieve
- **Action:** Schema with:
  - **Preconditions:** Must be true to apply action
  - **Add List:** Atoms that become true
  - **Delete List:** Atoms that become false

Closed World Assumption (CWA):
Any atom not explicitly stated is **false**.

Action Application:

$$s' = (s - \text{Del}(a)) \cup \text{Add}(a)$$

New state = old state minus deletions plus additions

PDDL Action Template:

```
(:action Move
 :parameters (?x ?from ?to)
 :precondition (and (At ?x ?from)
                    (Clear ?to))
 :effect (and (At ?x ?to)
              (Clear ?from)
              (not (At ?x ?from))
              (not (Clear ?to))))
```

Planning Strategies:

- **Forward Search (Progression):** Start state → Goal
- **Backward Search (Regression):** Goal → Start state
- **Heuristics:** Relaxed problem (ignore delete lists)

5. GAME THEORY

Normal Form Game:

- **Players:** $N = \{1, 2, ..., n\}$
- **Strategies:** S_i for each player i
- **Payoff:** $u_i(s_1, ..., s_n)$ for player i

Payoff Matrix (2-player):

Row \ Col	L	R
U	(u_1, u_2)	(u_1, u_2)
D	(u_1, u_2)	(u_1, u_2)

First number = Row's payoff, Second = Column's payoff

Strictly Dominated Strategy:

$$s_i \text{ dominated by } s_i' \iff u_i(s_i', s_{-i}) > u_i(s_i, s_{-i})\ \forall s_{-i}$$

Strategy that is ALWAYS worse than another

IESDS (Iterated Elimination):
Repeatedly eliminate strictly dominated strategies until no more can be removed.

NASH EQUILIBRIUM

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in S_i, \forall i$$

Definition: Strategy profile where NO player can improve by unilaterally changing their strategy.

Finding NE (Best Response Method):

1. For each column: underline Row's best payoff
2. For each row: underline Column's best payoff
3. Cells with BOTH underlined = Nash Equilibrium

Key Game Types:

- **Prisoner's Dilemma:** Defect dominates, but mutual cooperation is better
- **Coordination Game:** Multiple NE, prefer to coordinate
- **Zero-Sum:** $u_1 + u_2 = 0$ (pure competition)

Pareto Optimality:
Outcome is Pareto optimal if no player can be made better off without making another worse off.

6. SEARCH & PROJECT CODE

A* Search:

$$f(n) = g(n) + h(n)$$

- $g(n)$ = actual cost from start to n
- $h(n)$ = heuristic estimate from n to goal
- $f(n)$ = estimated total path cost through n

Heuristic Properties:

- **Admissible:** $h(n) \leq h^*(n)$ (never overestimates)
- **Consistent:** $h(n) \leq c(n, a, n') + h(n')$

Manhattan Distance (4-connected grid):

$$h(n) = |x_n - x_g| + |y_n - y_g|$$

Sum of horizontal + vertical distances. Admissible for 4-connected grids.

Search Algorithms:

- **BFS:** FIFO queue, optimal for unit costs
- **UCS:** Priority by $g(n)$, optimal
- **A*:** Priority by $f(n) = g(n) + h(n)$, optimal if h admissible

Bayesian Belief Update:

$$P(\text{obs}|E) = \frac{P(E|\text{obs}) \cdot P(\text{obs})}{P(E)}$$

Quick Reference: Essential Formulas

Bayes: $\frac{P(E|H)P(H)}{P(E)}$ **EU:** $\sum_s P(s)U(a,s)$ **Bellman:** $\max_a[R + \gamma \sum PV^*]$ **A*:** $f = g + h$